CreatorFlow AI

Software Requirements Specification

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1. Introduction

1.1 Project Overview

CreatorFlow AI is a full-stack, web-based platform that leverages large language model APIs to help digital creators generate optimized content metadata. The application targets creators operating on YouTube, TikTok, Instagram, and related platforms, providing AI-assisted tools for title generation, description writing, hashtag research, and SEO keyword recommendations — all from a single, unified dashboard.

The system is built on a modern JavaScript stack: a React frontend hosted on Vercel, a Node.js + Express backend hosted on Render, Firebase for authentication and data persistence, and the OpenAI API for AI-powered generation.

1.2 Purpose of This Document

This Software Requirements Specification (SRS) defines the complete set of functional, non-functional, interface, data, integration, testing, deployment, and maintenance requirements for CreatorFlow AI. It serves as the authoritative reference for all development, testing, and delivery activities and ensures alignment between stakeholders, the engineering team, and project reviewers.

1.3 Project Objectives

- Deliver a production-deployed web application within a 12-week, 6-sprint Agile Scrum cycle
- Implement secure user authentication and account management via Firebase Auth
- Integrate the OpenAI API to power four core AI generation tools: titles, descriptions, hashtags, and SEO optimization
- Provide a centralized dashboard for creating, saving, and managing generated content projects
- Produce a polished, portfolio-quality codebase hosted publicly on GitHub

1.4 Scope

In Scope — MVP

User authentication (sign-up, login, logout)
AI Title Generator, AI Description Generator, AI Hashtag Generator, AI SEO Optimizer
Project Save System (store, retrieve, and manage generated outputs)
Responsive dashboard UI (desktop and mobile browsers)
Cloud deployment on Vercel (frontend) and Render (backend)

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Out of Scope — Not Included in v1.0

Custom AI model training or fine-tuning
Native iOS or Android applications
Payment processing or subscription billing
Real-time multi-user collaboration
Direct social media publishing or feed integration
Advanced video analysis or computer vision features

1.5 Definitions & Abbreviations

Term	Definition
SRS	Software Requirements Specification — this document
MVP	Minimum Viable Product — smallest complete feature set delivering core value
API	Application Programming Interface — a contract for communication between software services
LLM	Large Language Model — an AI model generating human-like text (e.g. GPT-4)
UI	User Interface — the visual layer the user interacts with
UX	User Experience — the overall quality of the user interaction
CRUD	Create, Read, Update, Delete — standard data operations
CI/CD	Continuous Integration / Continuous Deployment — automated build and release pipelines
SEO	Search Engine Optimization — improving content discoverability and search rankings
NFR	Non-Functional Requirement — quality attributes such as performance and security

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2. Requirements Document

2a. Functional Requirements

Functional requirements define the specific behaviors, features, and capabilities the system must deliver. Each requirement below is labeled with a unique identifier for traceability to user stories and sprint tasks.

FR-01 User Authentication & Account Management

ID	Requirement	Priority
FR-01.1	The system shall allow new users to register with a valid email address and password	Must Have
FR-01.2	The system shall validate email format and enforce a minimum password length of 8 characters	Must Have
FR-01.3	The system shall authenticate registered users via Firebase Auth on login	Must Have
FR-01.4	The system shall display descriptive error messages for invalid credentials	Must Have
FR-01.5	The system shall maintain authenticated session state across page refreshes	Must Have
FR-01.6	The system shall allow users to log out, clearing the session and redirecting to the login page	Must Have
FR-01.7	All protected routes shall redirect unauthenticated users to the login page	Must Have

FR-02 Creator Dashboard

ID	Requirement	Priority
FR-02.1	The dashboard shall display navigation links to all four AI generation tools	Must Have
FR-02.2	The dashboard shall display the user's saved projects and recent generation history	Must Have
FR-02.3	The dashboard shall be accessible only to authenticated users	Must Have
FR-02.4	The dashboard shall render correctly on desktop and mobile viewport sizes	Should Have

FR-03 AI Title Generator

ID	Requirement	Priority
FR-03.1	The system shall accept user input for video topic, content tone, and target platform	Must Have
FR-03.2	The system shall submit the inputs to the OpenAI API and return 3–5 optimized title suggestions	Must Have
FR-03.3	Generated titles shall be displayed in clearly separated result cards	Must Have

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ID	Requirement	Priority
FR-03.4	Each title result shall include a one-click copy-to-clipboard action	Must Have
FR-03.5	Each title result shall include a save action that persists the output to the user's project	Must Have
FR-03.6	The system shall display a loading state while awaiting the API response	Should Have

FR-04 AI Description Generator

ID	Requirement	Priority
FR-04.1	The system shall accept user input for content topic and target platform	Must Have
FR-04.2	The system shall return a complete, platform-specific description incorporating relevant keywords	Must Have
FR-04.3	The description output shall include a copy-to-clipboard and save action	Must Have
FR-04.4	The system shall display a loading indicator during API processing	Should Have

FR-05 AI Hashtag Generator

ID	Requirement	Priority
FR-05.1	The system shall accept a content topic or video description as input	Must Have
FR-05.2	The system shall return 10–20 platform-relevant hashtags per request	Must Have
FR-05.3	Generated hashtags shall be displayed as a clearly formatted, copyable list	Must Have
FR-05.4	The user shall be able to copy all hashtags at once or save them to a project	Must Have

FR-06 AI SEO Optimizer

ID	Requirement	Priority
FR-06.1	The system shall accept a video topic or content idea as input	Must Have
FR-06.2	The system shall return SEO keywords and actionable optimization recommendations	Must Have
FR-06.3	Recommendations shall be displayed in a clear, structured format	Must Have
FR-06.4	The user shall be able to copy or save generated SEO suggestions to a project	Must Have

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FR-07 Project Save System

ID	Requirement	Priority
FR-07.1	The system shall allow users to save any generated output to a named project	Must Have
FR-07.2	Saved projects shall persist in Firebase Firestore and survive session termination	Must Have
FR-07.3	Users shall be able to view all saved projects from the dashboard	Must Have
FR-07.4	Users shall be able to delete individual saved items	Must Have
FR-07.5	Saved items shall display the generation type, content, and timestamp	Should Have

User Stories

The following user stories define acceptance criteria for all MVP features and are directly traceable to functional requirements and sprint deliverables.

ID	User Story	Acceptance Criteria
US-01	As a new user, I want to create an account so that I can securely access the platform.	User can register with email and password. Account is created in Firebase. User is redirected to dashboard on success. Error messages are displayed for invalid inputs.
US-02	As a returning user, I want to log in to my account so that I can access my saved projects.	User can log in with registered credentials. Session persists across page refreshes. Invalid credentials display appropriate error. Logout clears the session.
US-03	As a content creator, I want AI-generated titles so that I can publish optimized, high-performing content.	User inputs a video topic, tone, and target platform. AI returns 3–5 optimized title suggestions. Results are displayed clearly. User can copy or save any title.
US-04	As a content creator, I want AI-generated hashtags so that I can improve content discoverability and audience reach on social media platforms.	User enters a content topic or video description. AI generates 10–20 relevant hashtags. Generated hashtags are displayed clearly. User can copy or save generated hashtags.
US-05	As a content creator, I want AI-generated descriptions so that I can produce SEO-optimized content summaries efficiently.	User provides a topic and target platform. AI returns a full platform-specific description. Output includes relevant keywords. User can copy or save the description.
US-06	As a content creator, I want AI-generated SEO optimization suggestions so that I can improve content discoverability and search visibility across online platforms.	User enters a video topic or content idea. AI generates SEO keywords and optimization recommendations. Recommendations are displayed clearly. User can copy or save generated SEO suggestions.
US-07	As a creator, I want to save my generated content so that I can revisit and reuse it across future projects.	User can save any generated output to a named project. Saved content persists in Firebase. User can view, edit labels, and delete saved items from the dashboard.

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2b. Non-Functional Requirements

Non-functional requirements define the quality attributes, performance benchmarks, and system constraints that govern how the application behaves rather than what it does.

Performance

ID	Requirement
NFR-01	AI generation requests shall complete and display results within 10 seconds under normal network conditions
NFR-02	Dashboard and page navigation shall render within 2 seconds on a standard broadband connection
NFR-03	The application shall support at least 50 concurrent users without degradation during development testing

Usability

ID	Requirement
NFR-04	The interface shall follow intuitive navigation patterns requiring no onboarding documentation for basic use
NFR-05	All form inputs shall include placeholder text, labels, and inline validation feedback
NFR-06	The application shall be fully functional on current versions of Chrome, Firefox, Safari, and Edge
NFR-07	The application shall be responsive and usable on screen widths from 375px (mobile) to 1920px (desktop)

Security

ID	Requirement
NFR-08	All API keys and credentials shall be stored as server-side environment variables and never exposed to the client
NFR-09	All authenticated API routes shall validate Firebase Auth tokens server-side before processing requests
NFR-10	Firestore security rules shall restrict data reads and writes to the authenticated owner of each document
NFR-11	The application shall enforce HTTPS for all client-server communication in production
NFR-12	User passwords shall be managed exclusively by Firebase Auth; plain-text passwords shall never be stored

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Reliability & Availability

ID	Requirement
NFR-13	The application shall target 99% uptime leveraging Vercel and Render managed hosting infrastructure
NFR-14	The system shall handle OpenAI API errors gracefully, displaying user-friendly error messages rather than raw stack traces
NFR-15	Firebase Firestore shall serve as the single source of truth for all persisted user and project data

Maintainability

ID	Requirement
NFR-16	All source code shall follow consistent formatting standards enforced via a linter (ESLint)
NFR-17	Frontend components shall be modular and reusable to support future feature additions
NFR-18	The GitHub repository shall include a README with setup instructions, environment variable documentation, and a live demo link
NFR-19	All environment-specific configuration shall use .env files; a .env.example shall document all required variables

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2c. User Interface Requirements

The following requirements govern the visual design, interaction patterns, and layout of the CreatorFlow AI application.

ID	Requirement
UIR-01	The application shall implement a persistent left-sidebar navigation layout on desktop viewports
UIR-02	The sidebar shall display navigation links to: Dashboard, Generate Titles, Generate Descriptions, Generate Hashtags, SEO Suggestions, and Saved Projects
UIR-03	The active navigation item shall be visually highlighted with a distinct background color
UIR-04	All AI generation tool pages shall follow a consistent three-section layout: input form, generate button, results area
UIR-05	Input forms shall include labeled fields, placeholder text, and dropdown selectors for platform and tone where applicable
UIR-06	Generated result items shall be displayed as cards with copy-to-clipboard and save-to-project actions
UIR-07	Loading states shall be shown with a spinner or skeleton loader while awaiting API responses
UIR-08	The color scheme shall use the primary palette: Navy (#1E3A5F), Blue (#2E75B6), Light Blue (#4A9FD4), and neutral grays
UIR-09	Typography shall use a sans-serif system font stack (Arial or equivalent) for readability
UIR-10	On mobile viewports, the sidebar shall collapse into a hamburger menu or bottom navigation bar

The following diagrams illustrate the primary UI structure and system architecture:

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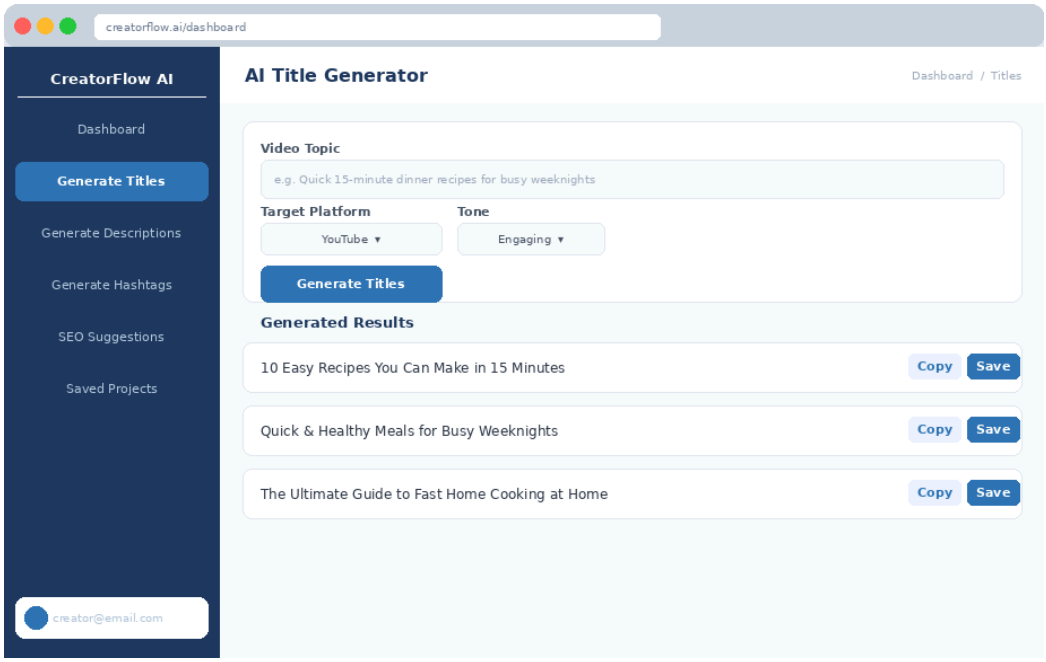


Figure 1 — CreatorFlow AI Dashboard UI Wireframe

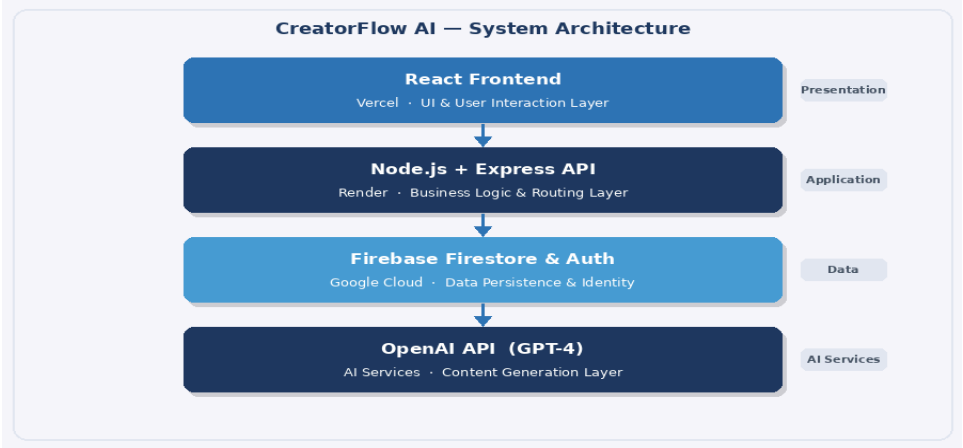


Figure 2 — CreatorFlow AI System Architecture

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2d. Data Requirements

This section defines the data entities, structures, flows, and storage strategy for CreatorFlow AI.

Data Entities

Entity	Fields	Storage
User	uid (Firebase), email, displayName, createdAt	Firebase Auth + Firestore
Project	projectId, userId (FK), name, createdAt, updatedAt	Firestore Collection: projects
GeneratedItem	itemId, projectId (FK), type (title/description/hashtag/seo), content, platform, createdAt	Firestore Subcollection: items
GenerationRequest	topic, platform, tone — transient request payload sent to OpenAI API	Not persisted (in-flight only)

Data Flow

Generation Request Flow

1. User submits input form (topic, platform, tone) in React frontend
2. Frontend sends authenticated POST request to Node.js backend API
3. Backend validates the Firebase Auth token in the request header
4. Backend constructs a prompt and calls the OpenAI API
5. OpenAI API returns generated content to the backend
6. Backend returns the result to the frontend
7. Frontend renders the result; user optionally saves to Firebase Firestore

Storage Requirements

ID	Requirement
DR-01	All user data shall be stored in Firebase Firestore using a document-based NoSQL model
DR-02	User projects shall be stored under a top-level 'projects' collection with userId as a query field
DR-03	Generated items shall be stored as a subcollection under their parent project document
DR-04	No sensitive user data (passwords, payment info) shall be stored in Firestore
DR-05	Firestore security rules shall ensure users can only read and write their own documents
DR-06	OpenAI API keys shall be stored only as server-side environment variables

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2e. Integration Requirements

CreatorFlow AI integrates with three external systems. Each integration is described below with its purpose, communication protocol, and error handling approach.

System	Purpose	Protocol	Error Handling
OpenAI API (GPT-4)	AI content generation for all four generation tools	HTTPS REST — POST /v1/chat/completions	Display user-friendly message; log error server-side; never expose API key to client
Firebase Auth	User registration, login, session management	Firebase SDK — client and server SDKs	Display Firebase error codes as human-readable messages
Firebase Firestore	Persistent storage for user projects and generated items	Firebase SDK — real-time document reads/writes	Retry on transient failures; display save error notifications

ID	Requirement
IR-01	All OpenAI API calls shall originate from the backend only; the API key shall never be sent to the client
IR-02	The backend shall validate the Firebase Auth ID token on every authenticated request before calling external APIs
IR-03	OpenAI prompt inputs shall be sanitized server-side to prevent prompt injection
IR-04	The application shall handle OpenAI rate limit errors (HTTP 429) gracefully with a user-facing retry message
IR-05	Firebase SDK versions shall be pinned in package.json to prevent breaking changes from automatic upgrades

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2f. Testing Requirements

A multi-layer testing strategy will be employed to validate all functional and non-functional requirements prior to deployment.

Testing Types

Type	Scope	Tools	Success Criteria
Unit Testing	Individual functions, utility helpers, and React components in isolation	Jest, React Testing Library	All unit tests pass with 80%+ code coverage on critical modules
Integration Testing	API routes, Firebase interactions, and OpenAI response handling	Jest, Supertest	All API endpoints return expected responses; Firebase read/write operations succeed
End-to-End Testing	Full user workflows: register, login, generate content, save to project, logout	Manual testing or Cypress	All 7 user stories pass acceptance criteria without errors
User Acceptance Testing	Simulated real-world usage sessions against all MVP features	Manual testing with structured test cases	A test user can complete all core workflows without documentation
Security Testing	Authentication enforcement, Firestore rules, environment variable exposure	Manual review + Firebase rules emulator	No unauthenticated route access; no API key visible in client network traffic
Performance Testing	AI generation response time, page load speed	Browser DevTools, Lighthouse	Generation < 10s; page load < 2s; Lighthouse score > 80

ID	Requirement
TR-01	All unit and integration tests shall be committed to the GitHub repository and run as part of the CI pipeline
TR-02	Each user story acceptance criterion shall have a corresponding manual test case documented before Sprint 6
TR-03	All critical bugs discovered in Sprint 6 testing shall be resolved before final deployment
TR-04	The application shall be tested on Microsoft Edge prior to submission
TR-05	Firebase Firestore security rules shall be validated using the Firebase Rules Emulator

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2g. Deployment Requirements

The following requirements govern the deployment environment, pipeline, and release process for CreatorFlow AI.

Component	Platform	Deployment Trigger	URL Type
React Frontend	Vercel (Hobby)	Push to main branch on GitHub (auto-deploy)	Production URL: creatorflow-ai-zeta.vercel.app
Node.js Backend	Render (Free Tier)	Push to main branch on GitHub (auto-deploy)	Production URL: creatorflow-api.onrender.com
Database & Auth	Firebase (Spark Free)	Managed — no deployment required	Firebase Console managed

ID	Requirement
DEPR-01	The frontend shall be deployed to Vercel with automatic deployments triggered on every push to the main branch
DEPR-02	The backend shall be deployed to Render with automatic deployments triggered on every push to the main branch
DEPR-03	All environment variables (API keys, Firebase config) shall be configured in the Vercel and Render dashboards, not in code
DEPR-04	The .env file shall be listed in .gitignore before the first commit is made
DEPR-05	A .env.example file shall document all required environment variable names without values
DEPR-06	The application shall be prepared for deployment by then end of Sprint 6, with live deployment completed at the developer’s discretion
DEPR-07	The GitHub repository README shall include setup instructions and local development configuration
DEPR-08	Incremental deployment shall begin no later than Sprint 3 to catch environment-specific issues early

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2h. Maintenance Requirements

Post-deployment maintenance requirements ensure the system remains functional, secure, and aligned with platform and API changes.

ID	Requirement
MR-01	All critical bugs identified post-deployment shall be triaged within 48 hours and resolved within one sprint cycle
MR-02	OpenAI API model versions shall be monitored for deprecation notices; model references in code shall be updated before end-of-life
MR-03	Firebase SDK and npm package dependencies shall be reviewed and updated on a monthly basis to address security vulnerabilities
MR-04	Firebase Firestore usage shall be monitored monthly against the free tier limits; upgrade paths shall be documented if usage grows
MR-05	Bug tracking and feature requests shall be managed at the developer's discretion using any preferred tracking method
MR-06	The README and technical documentation shall be updated to reflect any post-launch changes to architecture or features
MR-07	Environment variable configurations on Vercel and Render shall be reviewed when rotating API keys or Firebase credentials

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3. Agile Approach

3.1 Agile Principles Applied

CreatorFlow AI is developed in accordance with the Agile Manifesto, prioritizing working software over comprehensive documentation, responding to change over following a rigid plan, and delivering value incrementally over a single large release. Each sprint produces a functional, testable deliverable that builds directly on the previous sprint's output.

3.2 Scrum Framework

Scrum Team Roles

Product Owner — Owns the product backlog, defines acceptance criteria, and prioritizes features by value delivered to the end user

Scrum Master — Facilitates sprint ceremonies, removes blockers, enforces Scrum process discipline

Development Team — Designs, builds, and tests all features; responsible for sprint backlog estimates and delivery

Sprint Ceremonies

Ceremony	Frequency	Purpose
Sprint Planning	Start of each sprint	Review backlog, select sprint items, break into tasks, estimate effort
Daily Stand-up	Daily (async-friendly)	3-question check-in: What did I do? What will I do? Any blockers?
Sprint Review	End of each sprint	Demo completed functionality against acceptance criteria; gather feedback
Sprint Retrospective	End of each sprint	Reflect on team process: what went well, what to improve, action items
Backlog Refinement	Mid-sprint	Groom upcoming backlog items, clarify requirements, add detail to stories

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3.3 Sprint Plan

Sprint	Weeks	Goal	Key Deliverables
Sprint 1	1–2	Foundation	GitHub repo, React scaffold, Tailwind setup, base layout, routing
Sprint 2	3–4	Authentication	Firebase setup, sign-up/login/logout, protected routes, user data model
Sprint 3	5–6	Backend API	Node.js + Express scaffold, REST routes, middleware, frontend-backend connection
Sprint 4	7–8	AI Generation	OpenAI integration: title, description, hashtag, and SEO optimization tools
Sprint 5	9–10	Project Saving	Firebase persistence, save/retrieve generated items, dashboard integration
Sprint 6	11–12	Ship It	E2E testing, bug fixes, UI polish, Vercel + Render deployment, documentation

3.4 Product Backlog Management

The product backlog is the single, ordered list of all work required to deliver CreatorFlow AI. It is a living artifact maintained and prioritized throughout the project lifecycle.

Priority (MoSCoW)	Items
Must Have	Authentication system, AI Title Generator, AI Description Generator, AI Hashtag Generator, AI SEO Optimizer, Project Save System
Should Have	Dashboard generation history, copy-to-clipboard, loading states, responsive layout
Could Have	Dark mode, TikTok caption generator, thumbnail prompt generator, export to CSV
Won't Have (v1.0)	Payment systems, mobile apps, live collaboration, AI model training, social media publishing

Each sprint backlog is assembled during Sprint Planning by pulling the highest-priority Must Have items not yet completed. No Could Have items enter the sprint backlog until all Must Have items are done.

3.5 Definition of Done

A user story or task is considered Done when:

Feature is implemented and functional in a feature branch
Code has been self-reviewed and meets style standards (ESLint passing)
At least basic manual testing has been performed against acceptance criteria
Feature branch is merged to main via a Pull Request
Deployment builds successfully on Vercel / Render (no broken builds on main)

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> Documentation is updated if the feature changes architecture or setup

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4. Appendix

4.1 Traceability Matrix

The following matrix maps user stories to functional requirements and the sprint in which they are delivered.

User Story	Functional Req.	Sprint
US-01 — User Registration	FR-01.1, FR-01.2, FR-01.3	Sprint 2
US-02 — User Login	FR-01.3, FR-01.4, FR-01.5, FR-01.6	Sprint 2
US-03 — AI Title Generator	FR-03.1 – FR-03.6	Sprint 4
US-04 — AI Description Generator	FR-04.1 – FR-04.4	Sprint 4
US-05 — AI Hashtag Generator	FR-05.1 – FR-05.4	Sprint 4
US-06 — AI SEO Optimizer	FR-06.1 – FR-06.4	Sprint 4
US-07 — Project Save System	FR-07.1 – FR-07.5	Sprint 5

4.2 References

Resource	URL
React Documentation	https://react.dev
Node.js Documentation	https://nodejs.org/docs
Firebase Documentation	https://firebase.google.com/docs
OpenAI API Documentation	https://platform.openai.com/docs
Tailwind CSS Documentation	https://tailwindcss.com/docs
Scrum Guide (Schwaber & Sutherland)	https://scrumguides.org
GitHub Documentation	https://docs.github.com
Vercel Documentation	https://vercel.com/docs
Render Documentation	https://render.com/docs

4.3 Document Revision History

Version	Date	Description
v1.0	May 2026	Initial SRS document — all sections completed and submitted for review

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